

2. System = 1 physical qudit with "q" internal levels.

$$\mathbb{C}^q = \text{span} \{ |0\rangle, |1\rangle, \dots, |q-1\rangle \}$$

$$X|j\rangle = |j+1 \pmod{q}\rangle$$

$$Z|j\rangle = e^{2\pi i j/q} |j\rangle$$

"Basically if you keep applying X you wrap around instead of reaching an edge."

periodic X direction + periodic Z direction $\sim$ torus

A) $q=8$; $S_1 = X^4$; $S_2 = Z^4$

(i) Check $[S_1, S_2] = 0$

$$S_1 S_2 = S_2 S_1$$

$$\Rightarrow X^4 Z^4 = (XZ)^4$$

Now, $XZ = ZX e^{-2\pi i/q}$

So, $X^4 Z^4 = (ZX e^{-2\pi i/q})^4$

$$\Rightarrow x^4 z^4 = z^4 x^4 \left(e^{-8\pi i / 8} \right) \leftarrow 1$$

$$\Rightarrow x^4 z^4 = z^4 x^4$$

$$\therefore [s_1, s_2] = 0.$$

(ii) we need to show that codespace is 2-dimensional
so every state should satisfy:

$$|\psi\rangle = \alpha |0_L\rangle + \beta |1_L\rangle$$

and $|0_L\rangle$ and $|1_L\rangle$ are linearly independent.

$$\therefore \langle 0_L | 1_L \rangle$$

$$= \frac{(\langle 0| + \langle 4|)(|2\rangle + |6\rangle)}{2}$$

$$= \frac{\langle 0|2\rangle + \langle 0|6\rangle + \langle 4|2\rangle + \langle 4|6\rangle}{2}$$

$$= 0$$

$\therefore$ they are linearly independent

we need to find what generic states look like using stabilizers.

$$X^4|j\rangle = |j+4 \pmod{8}\rangle$$

$$X^4|0\rangle = |4\rangle; X^4|1\rangle = |5\rangle \dots$$

Let's consider $|\psi\rangle = C_0|0\rangle + C_1|1\rangle + C_2|2\rangle \dots + C_7|7\rangle$

$$X^4|\psi\rangle = C_0|4\rangle + C_1|5\rangle + C_2|6\rangle + C_3|7\rangle + C_4|0\rangle \\ + C_5|1\rangle + C_6|2\rangle + C_7|3\rangle$$

$$\text{Now, } Z^4|j\rangle = \left(e^{2\pi i j/q} |j\rangle\right)^4 \\ = e^{\pi i j} |j\rangle$$

Now, a stabilizer will always keep the state in the $+1$ eigen-state, so.

j needs to be even.

$$Z^4|j\rangle = |j\rangle \quad [j = 0, 2, 4, 6]$$

$$\therefore Z^4|\psi\rangle = C_0|0\rangle + C_2|2\rangle + C_4|4\rangle + C_6|6\rangle$$

Combining the effect of both the stabilizers.

$$X^4|\psi\rangle + Z^4|\psi\rangle = C_0(|0\rangle + |4\rangle) + C_2(|2\rangle + |6\rangle)$$

$$2|\psi\rangle = C_0(|0\rangle + |4\rangle) + C_2(|2\rangle + |6\rangle)$$

$$|\psi\rangle = \frac{C_0}{2}(|0\rangle + |4\rangle) + \frac{C_2}{2}(|2\rangle + |6\rangle)$$

which can be written down as,

$$|\psi\rangle = \frac{\alpha}{\sqrt{2}}(|0\rangle + |4\rangle) + \frac{\beta}{\sqrt{2}}(|2\rangle + |6\rangle)$$

(iii) Find X_L and Z_L s.t. $X_L^2 P = Z_L^2 P = P$.

$$X^2|\psi\rangle = \frac{\alpha}{\sqrt{2}}(|2\rangle + |6\rangle) + \frac{\beta}{\sqrt{2}}(|0\rangle + |4\rangle)$$

applying
the
stabilizer on
the errored
state

$$X^4(X^2|\psi\rangle) = X^6|\psi\rangle$$

$$= \frac{\alpha}{\sqrt{2}}(|0\rangle + |4\rangle) + \frac{\beta}{\sqrt{2}}(|2\rangle + |6\rangle)$$

$$= |\psi\rangle$$

$$X_L = X^2$$

$$|\psi\rangle = \frac{\alpha}{\sqrt{2}} (|0\rangle + |4\rangle) + \frac{\beta}{\sqrt{2}} (|2\rangle + |6\rangle)$$

$$= Z_L |0_L\rangle = |0_L\rangle; \quad Z_L |1_L\rangle = -|1_L\rangle$$

$$Z^K |0_L\rangle = e^{\pi i j k / 4} \cdot |0_L\rangle; \quad Z^K |1_L\rangle = e^{\pi i j k / 4} |1_L\rangle$$

for $j = 0, 4$ for $j = 2, 6$

↘ $K = \text{even}$ ↘ $K = 2$

$$\therefore \{X_L, Z_L\} = \{X^2, Z^2\}$$

$$(iii) \quad [X_L, S_1] \Rightarrow X^2 \cdot X^4 = X^4 \cdot X^2 = X^6$$

$$[X_L, S_2] \Rightarrow X^2 \cdot Z^4 = (X^2 \cdot Z^2) Z^2$$

$$= (Z X e^{-\pi i / 4})^2 \cdot Z^2$$

$$= Z^2 X^2 \cdot Z^2$$

$$= Z^2 (XZ)^2$$

$$= Z^2 (Z X e^{-\pi i / 4})^2$$

$$= -Z^2 Z^2 \cdot X^2 = Z^4 \cdot X^2$$

$$\begin{aligned}
[z_L, s_1] &\Rightarrow z^2 \cdot x^4 \\
&= (zx)^2 x^2 = (xz e^{-2\pi i/q})^2 x^2 \\
&= (-1xz)^2 \cdot x^2 \\
&= -1 x^2 z^2 x^2 \\
&= -1 \cdot -1 x^2 \cdot x^2 \cdot z^2 = x^4 \cdot z^2
\end{aligned}$$

$$\therefore [z_L, s_1] = 0$$

$$[z_L, s_2] \Rightarrow z^2 \cdot z^4 = z^4 \cdot z^2 = z^6$$

$$\therefore [z_L, s_2] = 0$$

... QED.